CHAPTER 11

Thermal Properties of Matter

Sr. No.	Concept	Formulas	Other information
1.	**Relation between different scale**	$\frac{\text{Reading on any scale} - \text{LFP}}{\text{UFP} - \text{LFP}} = \text{Constant for all scales}$ $\frac{C-0}{100} = \frac{F-32}{212-32} = \frac{K-273}{373-273} = \frac{\text{A} - \text{LFP}}{\text{UFP} - \text{LFP}}$ $\frac{C}{5} = \frac{F-32}{9} = \frac{K-273}{5} = \frac{\text{A} - \text{LFP}}{\text{UFP} - \text{LFP}}$ Difference in temperature: $\frac{\Delta C}{5} = \frac{\Delta F}{9} = \frac{\Delta K}{5}$ Celsius and Kelvin scales: $(T_2 - T_1)°C = (T_2 - T_1)K \quad \Rightarrow \quad \Delta T°C = \Delta TK$	C = temperature in centigrade scale F = temperature in the fahrenheit scale K = temperature in kelvin scale A = temperature in any other scale. T_2 = final temperature T_1 = initial temperature. LFP = lower fixed point. UFP = upper fixed point.
2.	**Linear expansion**	Final length $l = l_0 + \alpha l_0 \Delta T$ Change in length, $\Delta l = \alpha l_0 \Delta T$ Coefficient of linear expansion $\alpha = \frac{\Delta l}{l_0 \Delta T}$	α = coefficient of linear expansion l_0 = initial length at temperature T_1. l = final length at temperature T_2. ΔT = change in temperature = $(T_2 - T_1)$.
3.	**Areal expansion**	Final area, $A = A_0(1 + \beta \Delta T)$ Change in area, $\Delta A = A_0 \beta \Delta T$ Co-efficient of areal expansion $\beta = \frac{\Delta A}{A_0 \Delta T}$	β = coefficient of superficial (or areal) expansion A_0 = initial area
4.	**Volume or cubical expansion**	Final volume, $V = V_0(1 + \gamma \Delta T)$ Change in volume, $\Delta V = V_0 \gamma \Delta T$ Co-efficient of cubical expansion $\gamma = \frac{\Delta V}{V_0 \Delta T}$	γ = coefficient of cubical expansion V_0 = initial volume

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5.	**Relation between α, β and γ**	Three expansion coefficients related as; $\frac{\alpha}{1}=\frac{\beta}{2}=\frac{\gamma}{3}$ or $\beta=2\alpha$ and $\gamma=3\alpha$	γ = coefficient of volume expansion β = coefficient of areal expansion α = coefficient of linear expansion
6.	**Change in time period of simple pendulum on heating**	Time period of pendulum, $T=2\pi\sqrt{\frac{l}{g}}$ Fractional change in time period = $\frac{\Delta T}{T}=\frac{1}{2}\frac{\Delta l}{l}=\frac{1}{2}\alpha\Delta\theta$ Change in time; $\Delta T=\frac{1}{2}\alpha\Delta\theta T$	$\Delta\theta$ = change in temperature. T = time period of pendulum. α = Co-efficient of linear expansion l = length of the pendulum string.
7.	**Thermal stress in a rigidly fixed rod**	Thermal strain $=\frac{\Delta L}{L}=\alpha\Delta T$ $\left[\text{As } \alpha=\frac{\Delta L}{L}\times\frac{1}{\Delta T}\right]$ Thermal stress $=Y\alpha\Delta T$ Force on the supports $F=YA\alpha\Delta T$	ΔL = change in length. L = initial length. Y = Young's modulus of elasticity.
8.	**Apparent expansion and real expansion (Liquid)**	Coefficient of apparent expansion (γ_a): $\gamma_a=\frac{\text{Apparent expansion in volume}}{\text{initial volume}\times\Delta T}=\frac{(\Delta V)_a}{V\times\Delta T}$ Coefficient of real expansion (γ_r): $\gamma_r=\frac{\text{Real increase in volume}}{\text{initial volume}\times\Delta T}=\frac{(\Delta V)_r}{V\times\Delta T}$ $\Delta V_{\text{apparent}}=\Delta V_{\text{real}}-\Delta V_{\text{vessel}}$	ΔT = Change in temperature ΔV_{vessel} = change in volume of vessel.
9.	**Variation of density with temperature**	Change in density: $\rho=\rho_0(1+\gamma\Delta T)$ $\frac{\Delta\rho}{\rho}=-\gamma\Delta T$	ρ = density at temperature (T) ρ_0 = density at 0°C γ = temperature coefficient of volume expansion
10.	**Mechanical equivalent of heat (*J*)**	$\frac{W}{Q}=\text{constant}=J \Rightarrow W=JQ$	W = work done. Q = amount of heat
11.	**Specific Heat**	Specific heat. $\therefore\ S=\frac{Q}{m\Delta T}\Rightarrow Q\propto m\Delta T\Rightarrow Q=mS\Delta T$	Q = total heat required. S = Specific Heat capacity m = mass of substance ΔT = Change in temperature.

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12.	**Heat capacity or thermal capacity**	Heat capacity $=\frac{Q}{\Delta T}$ Heat capacity = mS = mass × specific heat	Q = total heat required. S = specific heat. ΔT = change in temperature.
13.	**Water equivalent**	Water equivalent of an object = Mass of the object × Specific heat of material of object or $W = MS$ H_C of substance $= H_C$ of water $M_sS_s = M_wS_w$ $M_sS_s = W \times 1$ cal g^{-1} °C^{-1}	W = water equivalent H_C = heat capacity M_s, M_w = mass of the substance and water respectively. S_s, S_w = specific heat of the substance and water respectively.
14.	**Molar specific heat**	$S_{\text{mol}} = \frac{M}{m}\left(\frac{Q}{\Delta T}\right) \quad \left[\therefore\ n = \frac{m}{M}\right]$ $S_{\text{mol}} = \frac{Q}{\mu\Delta T}$	M = molecular mass of the substance. m = mass of the substance n = number of moles
15.	**Latent heat**	Latent heat of substance. $L = \frac{Q}{m} \Rightarrow Q = mL$	m = mass of substance. L = latent heat of substance. Q = amount of heat required to change the state.
16.	**Principle of calorimetry**	Heat lost = Heat gained $m_1s_1\Delta T_1 = m_2s_2\Delta T_2$ $m_1s_1(T_1 - T_{\text{mix}}) = m_2s_2(T_{\text{mix}} - T_2)$ $(T_1 > T_{\text{mix}} > T_2)$	m_1, m_2 = mass of the two liquids s_1, s_2 = specific heat of two liquids. T_{mix} = final temperature of mixture. T_1, T_2 = initial temperature of two liquids
17.	**Heat current**	In steady state; $i_T = \frac{Q}{t} = \frac{KA(T_1 - T_2)}{L}$	$T_1 - T_2$ = temperature difference between two ends of the rod t = time L = length of rod. A = cross-sectional area K = thermal conductivity Q = total heat flown i_T = rate of heat flow

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18.	**Thermal resistance (R_T)**	$R_T = \frac{T_1 - T_2}{i_T} = \frac{T_1 - T_2}{KA(T_1 - T_2)/L}$ $R_T = \frac{L}{KA}$	K = thermal conductivity A= cross-section area of the rod L = length of the rod
19.	**Newton's law of cooling**	Rate of fall of temperature: $-\frac{dT}{dt} = K(T - T_0)$ If the temperature of body changes from T_1 to T_2 in time t. $\Rightarrow \left[\frac{T_1 - T_2}{t}\right] = K\left[\frac{T_1 + T_2}{2} - T_0\right]$	T = temperature of body T_0 = temperature of surrounding. T-T_0 = difference of temperature K = constant for any liquid. t = time
20.	**Reflectance, absorptance and transmittance**	$r + a + t = 1$ if $t = 0$, $a = 1 - r$	a = absorbing power of body. r = reflecting power of body. t = transmitting power of body.
21.	**Kirchoff's Law**	$\frac{e_\lambda}{a_\lambda} = E_\lambda =$ constant $\Rightarrow \left[\frac{e_\lambda}{a_\lambda}\right]_1 = \left[\frac{e_\lambda}{a_\lambda}\right]_2 =$ constant Hence $e_\lambda \propto a_\lambda$	e_λ = emissive power a_λ = absorptive power E_λ = emissive power of an ideal black body at that temperature.
22.	**Stefan's Law**	Heat energy emitted per second per unit area by a perfectly black body: $E \propto T^4 \Rightarrow E = \sigma T^4$ For bodies which are not perfectly black: $E = \varepsilon\sigma T^4$ If a body at temperature T is surrounded by body at temperature T_0, then Net rate of loss of energy per unit area from body surface is $E = E_1 - E_2 = \varepsilon\sigma T^4 - \varepsilon\sigma T_0^4 = \varepsilon\sigma\left(T^4 - T_0^4\right)$	σ = Stefan's constant = 5.67×10^{-8} watt m^{-2} K^{-4} (universal constant) T = absolute temperature of body T_0 = temperature of surrounded body. ε = emissivity of body.
22.	**Wein's displacement law**	The wavelength (λ_m) corresponding to maximum energy emission; $\lambda_m \propto \frac{1}{T} \Rightarrow \lambda_m = \frac{b}{T}$	b = Wein's constant = 2.89×10^{-3} mK) T = absolute temperature